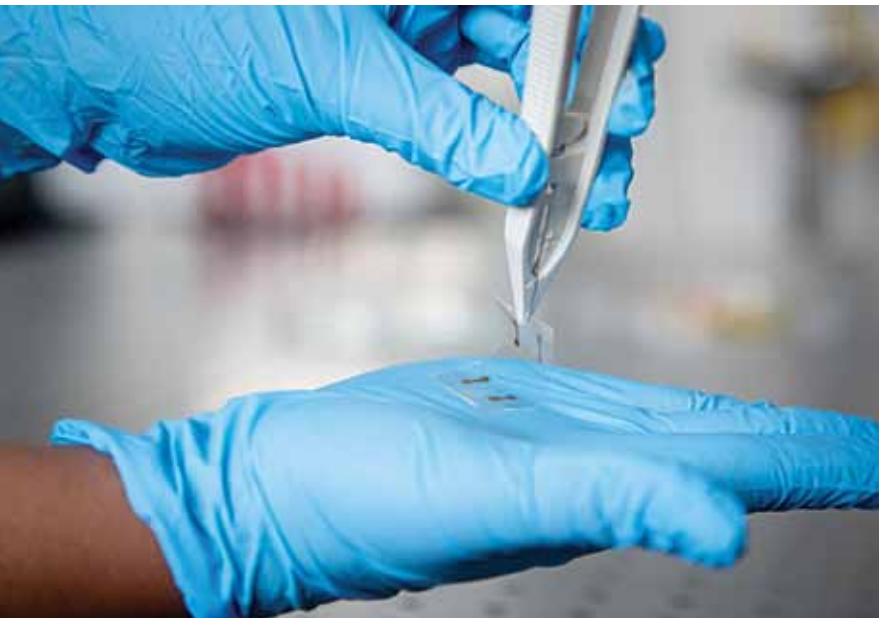




Next-gen transparent electronics for applications of the future



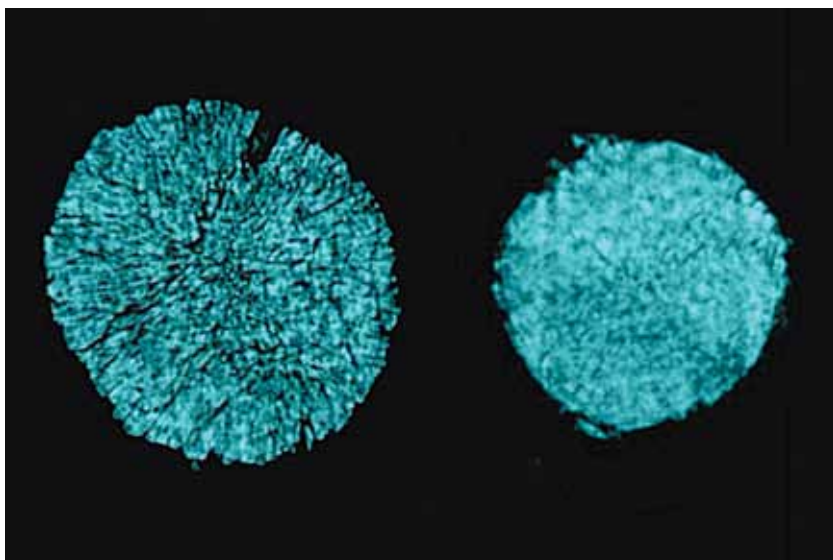
See-through devices could potentially be integrated in glass, flexible displays, and in smart contact lenses, bringing products closer to the future. Oxide based devices could also find use in power electronics and communication technology. An RMIT-led team has now developed an ultra-thin beta-tellurite semiconducting material family, which provides answers to high mobility p-type oxide, enabling fast, transparent circuits. Other key advantages include air stability, less stringent purity requirements, low cost, and easy deposition. To assess electronic properties, field-effect transistors (FETs) were fabricated. Thus, it was established that beta-tellurite is ten to one hundred times faster than existing p-type oxide semiconductors.

The optical transparency of the new materials could enable futuristic, flexible, transparent electronics (Credit: www.fleet.org.au)



Long lasting, power lithium batteries

Lithium-ion batteries have played a huge role in the development of light-weight and portable electronic devices. To improve the performance of existing devices and potentially enable new applications, a team of researchers at MIT and elsewhere has found a novel electrolyte that overcomes the problems of unwanted chemical reactions and enables better power-per-weight of next-generation batteries, without sacrificing the cycle life. The finding could make it possible for lithium-ion batteries to store about 420 watt-hours per kilogram (compared to the present capacity of 260 watt-hours per kilogram). That would translate into longer ranges for electric cars and longer-lasting changes on portable devices.



X-ray tomography images taken at Brookhaven National Lab show cracking of a particle in one electrode of a battery cell that used a conventional electrolyte (as seen on the left). The researchers found that using a novel electrolyte prevented most of this cracking (right) (Credit: <https://news.mit.edu>)